Chih-Yuan Chiu

Academic Employment

Georgia Institute of Technology — Aug 2024 - now
Research Engineer II,
School of Electrical and Computer Engineering.
Supervisor: Dr. Justin Romberg

Education

University of California, Berkeley — Sept 2018 - Dec 2023
Ph.D., Department of Electrical Engineering and Computer Sciences
Advisor: Dr. Shankar Sastry

National Taiwan University — Sept 2014 - June 2018
B.S., Department of Electrical Engineering

Contact Information

Emails: cyc@gatech.edu, chihyuanfrankchiu@gmail.com.
Website: https://chihyuanchiu.github.io/
Google Scholar: https://scholar.google.com/citations?hl=en&user=cl9ModoAAAAJ

Journal Publications

Jihoon Hong, **Chih-Yuan Chiu**, Sara Fridovich-Keil, Glen Chou. "PolyMerge: Compressing 3D Gaussian Splats with Polytope Coverings for Provably Safe Resource-Constrained Navigation." (Accepted at: *IEEE Robotics and Automation Letters (RA-L)*.)

Zhouyu Zhang⋆[1], **Chih-Yuan Chiu**⋆, Glen Chou. "Constraint Learning in Multi-Agent Dynamic Games from Demonstrations of Local Nash Interactions," vol. 11, no. 6, pp. 6696-6703, June 2026.

Chih-Yuan Chiu⋆, Zhouyu Zhang⋆, Glen Chou. "Learning Constraints from Stochastic Partially-Observed Closed-Loop Demonstrations," *IEEE Control Systems Letters (L-CSS)*, vol. 10, pp. 43-48, January 2026.

Chih-Yuan Chiu, Bryce Ferguson. "Robustness of Incentive Mechanisms Against System Misspecification in Congestion Games," in *IEEE Control Systems Letters (L-CSS)*, vol. 9, pp. 276-281, May 2025.

Chih-Yuan Chiu⋆, Jingqi Li⋆, Maulik Bhatt, Negar Mehr. "To What Extent Do Open-Loop and Feedback Nash Equilibria Diverge in General-Sum Linear Quadratic Dynamic Games?" *IEEE Control Systems Letters (L-CSS)*, vol. 8, pp. 2583-2588, November 2024.

Lasse Peters, Andrea Bajcsy, **Chih-Yuan Chiu**, David Fridovich-Keil, Forrest Laine, Laura Ferranti, Javier Alonso-Mora. "Contingency Games for Multi-Agent Interaction," *IEEE Robotics and Automation Letters (RA-L)*, vol. 9, pp. 2208-2215, January 2024. (*Best Paper Award Finalist: TC on Robot Control Best Paper Award*)

Forrest Laine, David Fridovich-Keil, **Chih-Yuan Chiu**, and Claire Tomlin. "The Computation of Approximate Generalized Feedback Nash Equilibria", *SIAM Journal on Optimization*, vol. 33, no. 1, pp. 294-318, 2023.

Druv Pai, Michael Psenka, **Chih-Yuan Chiu**, Manxi Wu, Edgar Dobriban, Yi Ma. "Pursuit of a Discriminative Representation for Multiple Subspaces via Sequential Games", *Journal of the Franklin Institute*, vol. 360, no. 6, pp. 4135-4171, April 2023.

Amay Saxena⋆, **Chih-Yuan Chiu**⋆, Ritika Shrivastava, Joseph Menke, Shankar Sastry. "Simultaneous Localization and Mapping: Through the Lens of Nonlinear Optimization," vol. 7, no. 3, *IEEE Robotics and Automation Letters (RA-L)*, July 2022.

[1] The asterisk superscript ⋆ indicates equal contribution.

Journal Submissions

Chih-Yuan Chiu, Devansh Jalota, Marco Pavone. "Credit-Based vs. Discount-Based Congestion Pricing: A Comparison Study," arXiv preprint 2602.11077, February 2026. URL: `https://arXiv.org/pdf/2602.11077`. (under review at: *Transportation Science*.)

Conference Publications

Dan BW Choe, Sundhar Vinodh Sangeetha, Steven Emanuel, **Chih-Yuan Chiu**, Samuel Coogan, Shreyas Kousik. "Ask, Reason, Assist: Robot Collaboration via Natural Language and Temporal Logic," arXiv preprint 2509.23506, September 2025. URL: `https://arXiv.org/pdf/2509.23506`. (Accepted at: *International Conference on Intelligent Robots and Systems (IROS) 2026*).

Tamer Başar[1⋆], Subhonmesh Bose[2⋆], Philip Brown[3⋆], **Chih-Yuan Chiu**[4⋆], Bryce L. Ferguson[5⋆], Chinmay Maheshwari[6⋆], Ketan Savla[7⋆], Raj Kiriti Velicheti[8⋆], Manxi Wu[9⋆]. "Control via Incentives and Information in Socio-technical Systems: A Tutorial." Accepted at: *2026 American Control Conference (ACC)*.

Shuyu Zhan⋆, **Chih-Yuan Chiu**⋆, Antoine Leeman, Glen Chou. "Robustly Constrained Dynamic Games for Uncertain Nonlinear Dynamics," arXiv preprint 2509.16826, September 2025. URL: `https://arXiv.org/abs/2509.16826`. Accepted at: *2026 IEEE International Conference on Robotics and Automation (ICRA)*.

Sundhar Vinodh Sangeetha, **Chih-Yuan Chiu**, Sarah H.Q. Li, Shreyas Kousik, "Language Conditioning Improves Accuracy of Aircraft Goal Prediction in Untowered Airspace," arXiv preprint 2509.14063, September 2025. URL: `https://arXiv.org/abs/2509.14063`. Accepted at: *2026 IEEE International Conference on Robotics and Automation (ICRA)*. (*Link to Media*).

Zheng Qiu, **Chih-Yuan Chiu**, Glen Chou. "Active Constraint Learning in High Dimensions from Demonstrations," arXiv preprint 2512.22757, December 2025. Accepted at: *8th Annual Learning for Dynamics & Control Conference (L4DC)*.

Chih-Yuan Chiu. "Approximately Optimal Toll Design for Efficiency and Equity in Arc-Based Traffic Assignment Models," *61st Annual Allerton Conference on Communication, Control, and Computing*, September 2025.

Chih-Yuan Chiu, Devansh Jalota, Marco Pavone. "Credit vs. Discount-Based Congestion Pricing: A Comparison Study," in *IEEE Conference on Decision and Control (CDC)*, pp. 2331-2336, December 2024.

Chih-Yuan Chiu, Shankar Sastry. "Parameter Estimation in Optimal Tolling for Traffic Networks Under the Markovian Traffic Equilibrium," in *American Control Conference (ACC)*, pp. 1461-1467, July 2024.

Chih-Yuan Chiu, Chinmay Maheshwari, Pan-Yang Su, Shankar Sastry. "Dynamic Tolling in Arc-based Traffic Assignment Models," in *58th Annual Allerton Conference on Communication, Control, and Computing*, pp. 1-8, September 2023.

Chih-Yuan Chiu⋆, Chinmay Maheshwari⋆, Pan-Yang Su, Shankar Sastry. "Arc-based Traffic Assignment: Equilibrium Characterization and Learning," in *IEEE Conference on Decision and Control (CDC)*, pp. 7751-7758, December 2023.

Jingqi Li, **Chih-Yuan Chiu**, Lasse Peters, Fernando Palafox, Mustafa Karabag, Javier Alonso-Mora, Somayeh Sojoudi, Claire Tomlin, David Fridovich-Keil. "Scenario-Game ADMM: A Parallelized Scenario-Based Solver for Stochastic Noncooperative Games," in *IEEE Conference on Decision and Control (CDC)*, pp. 8093-8099, December 2023.

Chih-Yuan Chiu. "SLAM Backends with Objects in Motion: A Unifying Framework and Tutorial," in *American Control Conference (ACC)*, pp. 1635-1642, May 2023.

Jingqi Li, **Chih-Yuan Chiu**, Lasse Peters, Somayeh Sojoudi, Claire Tomlin, David Fridovich-Keil. "Cost Inference for Feedback Dynamic Games from Noisy Partial State Observations and Incomplete Trajectories," in *International Conference on Autonomous Agents and Multiagent Systems*, pp. 1062-1070, June 2023.

Chih-Yuan Chiu, David Fridovich-Keil. "GTP-SLAM: Game-Theoretic Priors for Simultaneous Localization and Mapping in Multi-Agent Scenarios," in *IEEE Conference on Decision and Control (CDC)*, pp. 247-252, December 2022.

Chinmay Maheshwari⋆, **Chih-Yuan Chiu⋆**, Eric Mazumdar, Shankar Sastry and Lillian J. Ratliff. "Zeroth-Order Methods for Convex-Concave Minmax Problems: Applications to Decision-Dependent Risk Minimization", in *25th International Conference on Artificial Intelligence and Statistics* (AISTATS), vol. 51, pp. 6702-6734, 2022.

Chih-Yuan Chiu⋆, David Fridovich-Keil⋆, and Claire Tomlin. "Encoding Defensive Driving as a Dynamic Nash Game," in *IEEE International Conference on Robotics and Automation (ICRA)*, pp. 10749-10756, 2021.

Forrest Laine, David Fridovich-Keil, **Chih-Yuan Chiu**, and Claire Tomlin. "Multi-Hypothesis Interactions in Game-Theoretic Motion Planning", in *IEEE International Conference on Robotics and Automation (ICRA)*, pp. 8016-8023, June 2021.

Forrest Laine, **Chih-Yuan Chiu**, Claire Tomlin, "Eyes-Closed Safety Kernels: Safety for Autonomous Systems Under Loss of Observability", in *Robotics: Science and Systems (RSS)*, July 2020.

Conference Submissions

Zhouyu Zhang, **Chih-Yuan Chiu**, and Glen Chou. "Learned Dynamics Enable Optimality Regularizers for Data-Efficient Offline Inverse Constrained RL." (under review at: *Annual Conference on Neural Information Processing Systems (NeurIPS) 2026*.)

Theses

(*Ph.D. Thesis*) **Chih-Yuan Chiu**. "Algorithm Design for Safe and Efficient Societal-Scale Navigation," 2023. (`https://www2.eecs.berkeley.edu/Pubs/TechRpts/2023/EECS-2023-267.pdf`).

(*Master's Thesis*) **Chih-Yuan Chiu**. "Simultaneous Localization and Mapping: A Rapprochement of Filtering and Optimization-Based Approaches," 2021. (`https://www2.eecs.berkeley.edu/Pubs/TechRpts/2021/EECS-2021-76.pdf`).

Invited Oral Presentations

3rd MAD-Games Workshop at the 2026 American Control Conference (ACC) May 2026
(*Upcoming*) *"Constraint Learning in Multi-Agent Dynamic Games from Demonstrations of Local Nash Interactions"*

11th Georgia Tech Postdoctoral Research Symposium Nov 2025
"Credit-based vs. Discount-based Congestion Pricing: A Comparison Study." (Top 10 Submissions)

Georgia Tech DCL Symposium Spotlight Talk Apr 2025
"Credit-based vs. Discount-based Congestion Pricing: A Study of Tolling and Toll Relief Mechanisms for Traffic Management."

Georgia Tech ECE Decision and Control Laboratory (DCL) Student Seminar Oct 2024
"Interaction-aware Multi-Agent Control: The Role of Information."

UC Berkeley EECS C106B/206B Guest Lecture Apr 2022
"Simultaneous Localization and Mapping: A Unifying Optimization-Based Framework." (with Ritika Srivastava)

UC Berkeley Semiautonomous Seminar Feb 2022
"Towards a Rapprochement of Estimation, Prediction, and Planning for Autonomous Navigation."

UC Berkeley EE 221A Guest Lecture Dec 2021
"Simultaneous Localization and Mapping: Filtering and Optimization Approaches." (with Amay Saxena)

UC Berkeley Semiautonomous Seminar Sep 2021
"Gradient Free Optimistic Gradient Descent Ascent: Applications to Decision Dependent Risk Minimization." (with Chinmay Maheshwari)

UC Berkeley Semiautonomous Seminar Jun 2021
"Factor Graphs: A Tool for Optimization-Based Inference in Robotics."

UC Berkeley EECS C106B/206B Guest Lecture Feb 2021
"Simultaneous Localization and Mapping: A Unifying Optimization-Based Framework." (with Amay Saxena)

UC Berkeley Semiautonomous Seminar Jun 2020
"Adversarial-to-Cooperative Games."

Invited Poster Presentations

2025 NSF FRR-NRI Annual Meeting Oct 2025
"Robustly Constrained Dynamic Games with Uncertain Dynamics."

Teaching Experience (Georgia Tech)

6-Lecture Short Course (Game Theory in Controls) Jul 2025 - Aug 2025
Instructor for a 6-lecture Short Course

Teaching Experience (UC Berkeley)

EE 127 (Optimization Models in Engineering) Aug 2023 - Dec 2023
20-hour Graduate Student Instructor for a 16-week semester-long course

EE 221A (Linear Systems Theory) Aug 2021 - Dec 2021
20-hour Graduate Student Instructor for a 16-week semester-long course

EE 16B (Designing Information Devices and Systems II) Jul 2020 - Aug 2020
25-hour Graduate Student Instructor for an 8-week summer course

Journal and Conference Reviewing

Journals
- IEEE Transactions on Human Machine Systems.
- IEEE Transactions on Networked Control Systems.
- IEEE Transactions on Automatic Control.
- IEEE Control Systems Letters.
- IEEE Robotics and Automation Letters (RA-L).
- Journal of Machine Learning Research (JMLR).

Conferences
- IEEE International Conference on Intelligent Transportation Systems (ITSC).

- IEEE Conference on Decision and Control (CDC).
- American Control Conference (ACC).
- IEEE International Conference on Robotics and Automation (ICRA).
- IEEE International Symposium on Multi-Robot and Multi-Agent Systems (MRS).

Software Skills Python, Matlab, C++, LaTeX

Research Grants (*Under Review*) Amazon Research: "Dynamic Closed-Loop Polytope-based Scene Representations for Sustained, Provably Safe, and Resource-Constrained Warehouse Navigation." Role: **PI**. $160,946.

(*Under Review*) Amazon Research: "Reliable Game-Theoretic Motion Planning with Robust Temporal Logic Constraint Satisfaction for Interactive Warehouse Robot Systems." Role: **PI**. $150,000.

(*Under Review*) NSF: "Reliable Synthesis and Learning Methods for Safe, Constrained Multi-Agent Interactions in Dynamic Games." Role: **Co-PI**. $600,000.

Service and Outreach

ACC 2026: Tutorial on Control via Incentives and Information in Socio-technical Systems, Co-Organizer May 2026
Tutorial on Control via Incentives and Information in Socio-technical Systems

ACC 2026: Workshop on Autonomy in Transportation, Co-Organizer May 2026
Workshop on Emerging Challenges in Multi-agent Planning and Control (https://sites.google.com/view/acc2026w05multiagentautonomy/home)

M.S. Thesis Defense Committee (Maria Cayetana Salinas Rodriguez, ECE) April 2026

M.S. Thesis Defense Committee (Sundhararajan Vinodh Sangeetha) April 2026

Graduate Student Recruitment Committee (GSRC) December 2025 - May 2026
Committee for evaluating Ph.D. applications to Georgia Tech ECE and recommending applicants for first-year Graduate Teaching Assistantships (GTA)

ACC 2025: Workshop on Mixed-Autonomy Traffic, Co-Organizer July 2025
Workshop on Emerging Challenges and Opportunities in Mixed Autonomy Transportation Systems (https://sites.google.com/view/acc2025workshoponmixedautonomy/home?authuser=0)

M.S. Thesis Proposal Committee (Maria Cayetana Salinas Rodriguez) December 2025

M.S. Thesis Proposal Committee (Sundhararajan Vinodh Sangeetha) November 2025

Decision and Control Laboratory Seminar Co-Organizer Sep 2024 - now
Regular seminar series focused on emerging challenges in the control of robotic and cyber-physical systems, at Georgia Tech ECE

Semiautonomous Seminar Co-Organizer Jan 2020 - Dec 2021
Weekly seminar on control theory, robotics, optimization, computer vision, and machine learning, at UC Berkeley EECS.